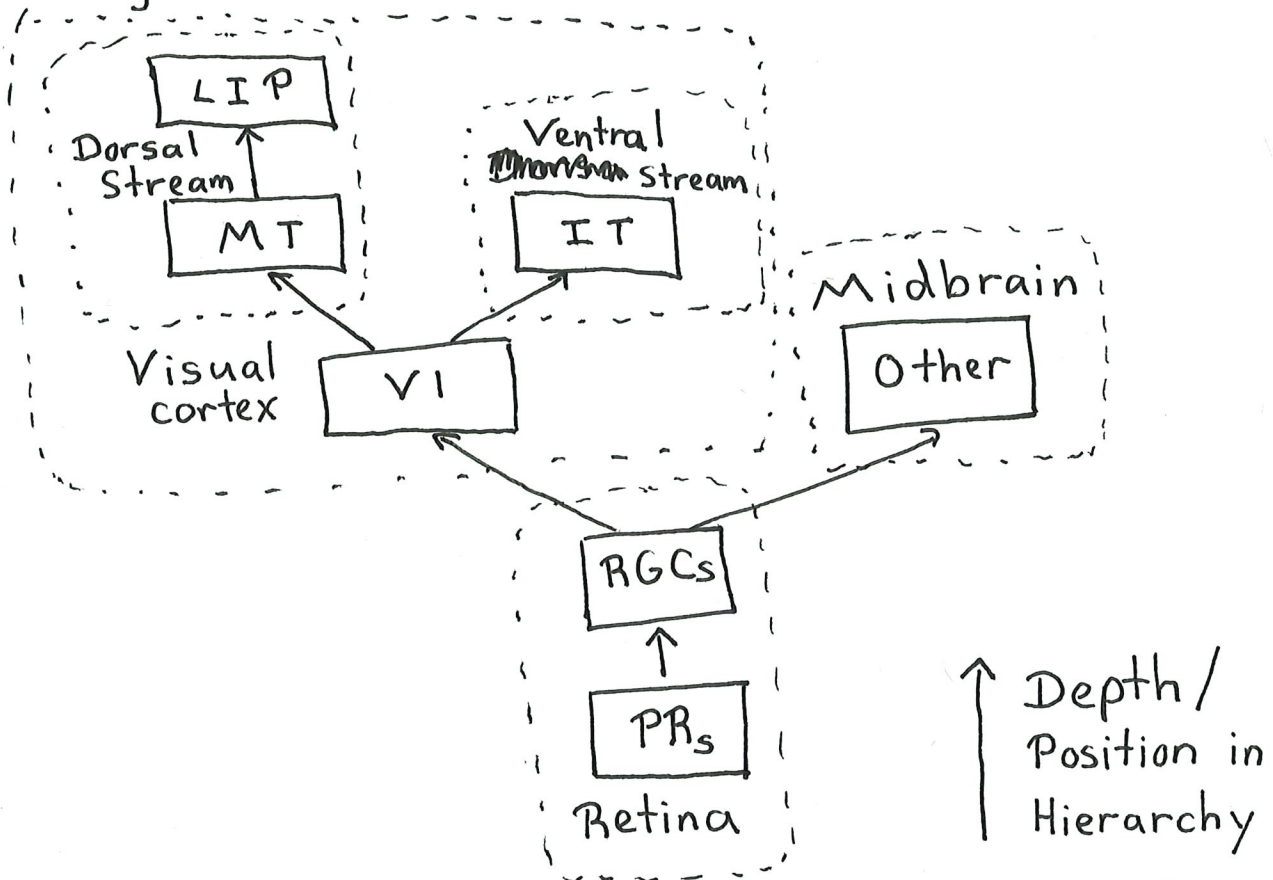
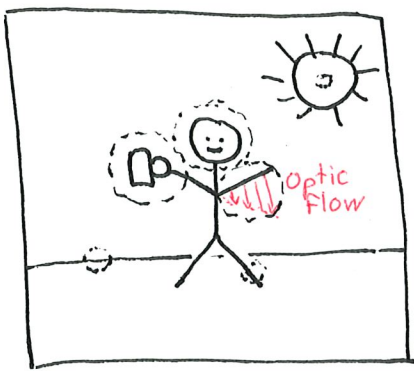


The computational organization of vision:

Key regions discussed in this class



What makes a visual neuron respond?



- What stimulus is shown: feature selectivity.
(e.g.: light, motion, objects)
- Where that stimulus is located: receptive field
(full coverage in each area)

Deeper in the hierarchy: receptive fields get bigger; history dependencies get longer; cells become ~~sensitive~~ ^{selective} to complex features; cells become invariant to certain details.

Example cells: "ON" and "OFF" light detectors in RGCs; motion detectors in RGCs, VI, MT; motion integrators in LIP; "object cells" and "Face cells" in IT; edge detectors in VI.

Linear-nonlinear-Poisson (LNP) models

Many sensory neurons can be modeled as:

$$k \sim \text{Poisson}(\mathcal{F}(L(s)))$$

Poisson sampling

s : stimulus

L : linear transformation

$\mathcal{F}$: nonlinear function

k : spike counts

For early visual neurons, L models the receptive field properties of the neuron, and s is visual contrast.

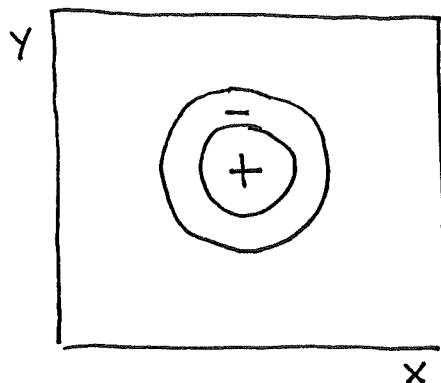
$$s(x, y, t) = \frac{I(x, y, t) - I_0}{I_0}$$

I : luminance of light

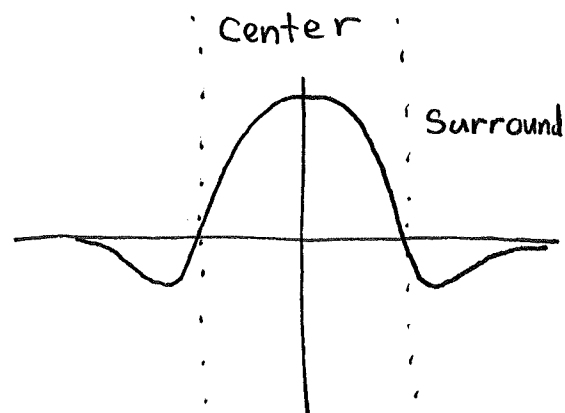
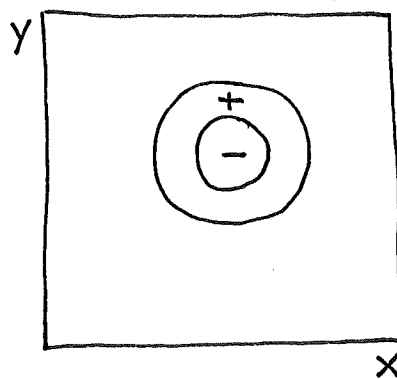
I_0 : average luminance

RGC example:

"ON" RGC



"OFF" RGC



Ignoring time for now:

$$L(s) = \int dx' \int dy' s(x', y') \underbrace{K(x_0 - x', y_0 - y')}_{\text{Receptive field kernel}}$$

Spatial convolution

$$K(x, y) = \underbrace{\pm \left(\frac{A}{2\pi\sigma_c^2} e^{-(x^2+y^2)/(2\sigma_c^2)} - \frac{B}{2\pi\sigma_s^2} e^{-(x^2+y^2)/(2\sigma_s^2)} \right)}_{\text{Difference of Gaussians receptive field}}$$

ON/OFF Center surround

For ON/OFF RGCs, S is well-modeled as threshold-linear.

$$h = [L(s)]_+ = \begin{cases} L(s), & L(s) \geq 0 \\ 0, & L(s) \leq 0 \end{cases}$$

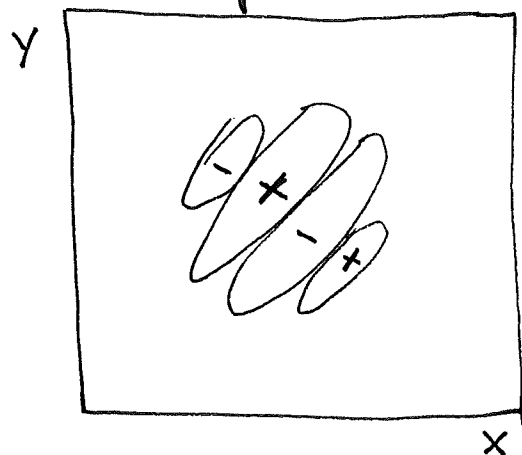
Since each population of RGCs covers the visual field, it can be convenient to index neurons by the location of their receptive field.

$$h_{x,y}^{(ON)} = [(K^{(ON)} * S)(x,y)]_+; h_{x,y}^{(OFF)} = [(K^{(OFF)} * S)(x,y)]_+$$

These "Feature maps" represent the visual world in two ways at once - "ON" signals & "OFF" signals.

"Feature maps" are also fundamental to convolutional neural networks in machine learning. In that setting, many K are learned with backprop, just like weights.

V1 Example:
"Simple" cell



Orientation tuning:

Stimulus	Response
Gratings	
	
	
	
	

Gabor kernel in vertical orientation:

$$K(x, y) = \frac{A}{2\pi\sigma_x\sigma_y} \underbrace{e^{-(x^2/(2\sigma_x^2) + y^2/(2\sigma_y^2))}}_{\text{Gaussian falloff}} \underbrace{\cos(kx + \phi)}_{\text{sinusoidal lobes}}$$

For orientation θ , substitute

$$x \rightarrow x \cos \theta + y \sin \theta$$

$$y \rightarrow y \cos \theta - x \sin \theta$$

Nonlinear part rectifies and saturates. Typical model

$$f(x) = k_{\max} \frac{[x]_+^2}{\alpha^2 + [x]_+^2} ; \quad \begin{array}{l} k_{\max}: \text{maximal firing rate} \\ \alpha: \text{half-maximum point} \end{array}$$

The resulting feature map is analogous to a "wavelet" transform in signal processing. This technique underlies JPEG compression, so this is a very efficient way to represent images.

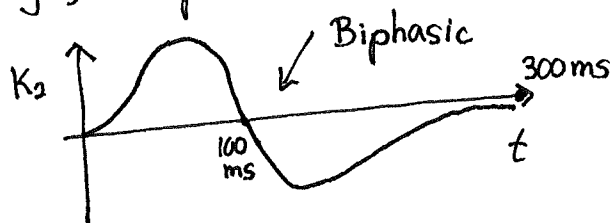
Spatiotemporal receptive fields:

When we include time,

$$L(s) = \underbrace{\int dx' \int dy' \int dt' s(x', y', t') K(x_0 - x', y_0 - y', t - t')}_{\text{Spatiotemporal convolution}}$$

Separable receptive field (e.g., simple cell):

$$K(x, y, t) = \underbrace{K_1(x, y)}_{\text{Spatial component}} \underbrace{K_2(t)}_{\text{Temporal component}}$$



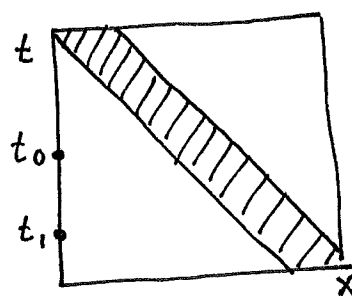
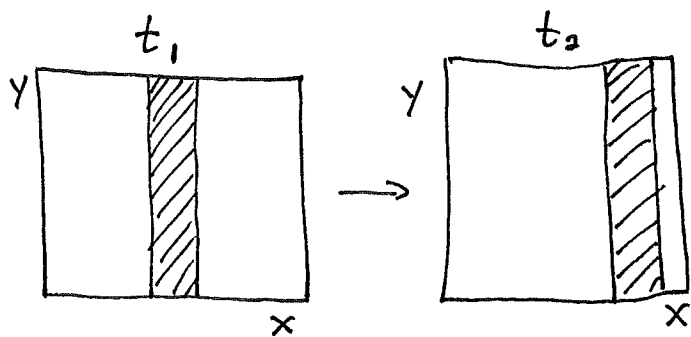
Little response to static image.

Non-separable receptive field (e.g., ON RGC):

Amplitudes of center & surround, A and B in previous equation, become time-dependent, $A(t)$ & $B(t)$. The center is typically faster and more biphasic than surround.

Direction-selectivity (e.g., RGCs, V1, MT)

Motion is orientation in spacetime.



"Spacetime" diagram.

Cells can become direction-selective by having Gabor-like spatiotemporal ~~the~~ receptive fields oriented in space and time. Preferred orientation $\Rightarrow$ preferred velocity.

For example, the following receptive field would lead a neuron to respond to the rightward motion shown above.

